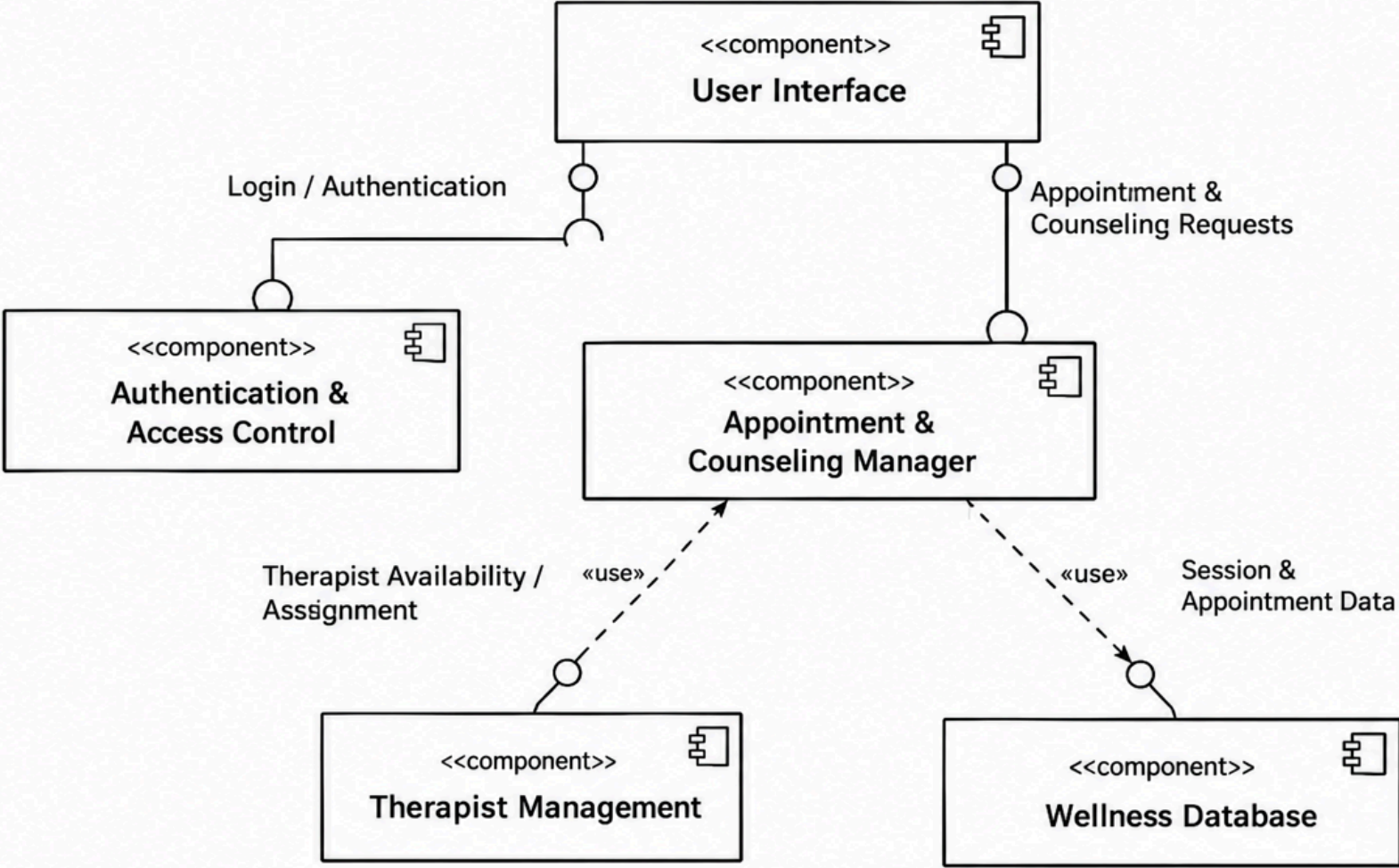


Corporate Mental Wellness & Counseling Platform - Component Diagram



Legend:

- <<component>>**  Component (UML notation)
-  Provided Interface (ball)
-  Required Interface (socket)
-  Assembly Connector
-  Usage Dependency (<<use>>)

Lab 3: Architecture Selection & Justification

Name: Rohan Rampura Praveen Kumar

SRN: PES1UG24CS382

Project: Corporate Mental Wellness & Counseling Platform

Domain: Healthcare & Telemedicine

Selected Architecture: Layered Architecture

Architecture Selection

The **Layered Architecture** is selected for the Corporate Mental Wellness & Counseling Platform because it provides a clear separation between the user interface, application logic, access control, therapist management, and data storage components. The architecture is suitable for organizing the platform into distinct responsibilities while keeping interactions between components clearly defined.

Reason 1 – Clear Separation of Responsibilities

The platform contains different functions such as user interaction, authentication, appointment and counseling management, therapist management, and wellness data storage. A layered structure separates these responsibilities into different components, making the system easier to understand, maintain, and modify. Changes to one functional area can be made with less impact on unrelated parts of the system.

Reason 2 – Maintainability and Modularity

The architecture supports modular development by keeping major system functions separated. For example, the **Authentication & Access Control** component can handle user authentication and access permissions, while the **Appointment & Counseling Manager** handles appointment and counseling requests. This separation makes individual components easier to test, update, and maintain.

Security Advantage

Layered separation provides a security advantage by allowing **authentication and access control to be handled as a dedicated component**. User access can be verified before requests reach protected application functions or wellness data. The

Wellness Database can therefore be accessed through controlled application components rather than directly from the user interface.

Performance Benefit

The separation of application functions allows frequently used operations such as authentication, appointment handling, and therapist availability management to be processed independently. This can reduce unnecessary processing between unrelated functions and allows the system to handle user requests through the appropriate component.

Conclusion

Layered Architecture provides a structured organization for the Corporate Mental Wellness & Counseling Platform. It separates user interaction, security, counseling management, therapist management, and data storage into distinct components, supporting **maintainability, modularity, security, and efficient processing**.